

Nishkarsh Jain

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Objective

Aspiring VLSI engineer seeking an Early Career opportunity. Through internships at Synopsys, Nvidia and Jabil, I have hands-on experience in Emulation, UART protocol, STA, Physical Design, Scripting and automation. I am committed to contributing to innovative projects by utilising my core knowledge and excellent problem-solving skills.

Education

THAPAR INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.E., Electronics and Communication Engineering. CGPA-9.05
(Merit Scholarship - III awardee)

Patiala, Punjab
2021 - 2025

SILVER BELLS PUBLIC SCHOOL

Senior Secondary Education(XII). Percentage-93.5%

Shamli, Uttar Pradesh
2018 - 2020

ST. FRANCIS SCHOOL

Secondary Education(X). Percentage-93.8%

Shamli, Uttar Pradesh
2018

Skills & Interests

Technical: Verilog, System Verilog, C/C++, PERL, Python, PCB design

Embedded Hardware: Adalm-Pluto SDR, KC705 FPGA board, STM32, ESP-32, Arduino, Sensors

Laboratory: Synopsys VCS, Synopsys Prime Time, Xilinx Vivado, Matlab, Simulink, S-Edit, Silvaco, Arduino IDE

Experience

Synopsys India Pvt. Ltd.

Technical Engineer Apprentice

Noida, Uttar Pradesh
September 2025 - **Ongoing**

- Feature Development and Bug fixes for UART Transactor of Zebu emulator using Verilog and C++
- Code coverage improvement, Valgrind fixes and Coverity fixes for multiple transactors
- Development of HW/SW Testbench Backwards Compatibility Agent
- Development and Implementation of complete workflow of Compatibility for Zebu Transactors with added AI review for every issue dissection using Python

Nvidia Graphics Pvt. Ltd.

ASIC Engineer Intern ([here](#))

Bangalore, Karnataka
January 2025 - July 2025

- Timing Closure and Performance Optimisation for Advanced SoC Designs using Static Timing Analysis (STA)
- Perl Scripts for Automating STA report analysis
- Manual fixes for clock transition and crosstalk violations

Jabil Circuit India Private Limited

Test Engineer Intern ([here](#))

Pune, Maharashtra
June 2024 - July 2024

- Working, Debugging and Testing of Ericsson's 5G Radio Transceiver - AIR3268
- SMT Process of Ericsson AIR 3268 Transceiver PCBA
- Setup, measurement, and calibration of Keysight test equipment

Academic Projects

Development and Implementation of a 5G Modem using a FPGA Board

January 2024 - January 2025

- FM Broadcast and Radio using Adalm-Pluto SDR [🔗](#)
- Hardware-in-the-Loop Simulation of FFT on Kintex-7 KC705 FPGA Development Board
- Simulink Model Development, FPGA implementation of 5G transceiver using Vitis Model Composer

Robosoccer Bot [🔗](#)

February 2024 - April 2024

- Built an RC-controlled bot using a laser-cut metal chassis, Sabertooth motor driver, and geared motors

Test Data Compression for System-on-a-Chip Using Huffman Codes

April 2024

- Implementation of Huffman Codes for data compression using Xilinx Vivado

Smart Irrigation System [🔗](#)

June 2023 - July 2023

- Worked on a smart irrigation system for flood-irrigated paddy fields using IoT-based systems as group leader.
- Used ESP-32 microcontroller, nRF24L01 RF module, GSM SIM800A, and BME280 & Thingspeak IoT Cloud.
- Worked on water conservation by automating water flow and alerting the farmer about the weather.

Certifications

ZEBU Emulator Fundamental Training

2025

VLSI Design-FPGA and ASIC ([here](#))

2023

IoT-Based Systems ([here](#))

2022 - 2023